

# Column sizing: the actual basis and what is still unqualified

**No complete column design has been accepted.** The effective-resistance transport model is authorized as a stated pre-pilot assumption. That approval does not waive numerical, stability or product-duty checks.

Feed: **4000 L/h** · Temperature: **25 °C** · Solvent/oil mass ratio: **0.5** · Wet-solvent water: **1.5 wt%**. Fresh RRBO NMP and water remain exactly zero.

| Requested output           | Current acceptance |
|----------------------------|--------------------|
| Active height              | Not qualified      |
| Physical compartment count | Not qualified      |
| Overall efficiency         | Not qualified      |
| Selected operating RPM     | Not qualified      |
| Selected diameter          | Not qualified      |

## The saved equilibrium trial does not meet the full duty

The recorded trial used 10 configured equilibrium stages. It is a reproduced physical-LLE trial, not an accepted theoretical stage count or a global limit on every possible process design.

| Property                      | Recorded/recomputed | Target | Basis           | Check |
|-------------------------------|---------------------|--------|-----------------|-------|
| Total aromatics, including PA | 7.5036              | ≤ 5    | wt% hydrocarbon | FAIL  |
| Polar aromatics (subset)      | 0.1542              | ≤ 0.5  | wt% hydrocarbon | PASS  |
| Hydrocarbon recovery          | 88.7061             | ≥ 83   | %               | PASS  |
| Saturates                     | 92.4964             | ≥ 90   | wt% hydrocarbon | PASS  |
| NMP in wet raffinate          | 6.3825              | ≤ 7    | wt% full phase  | PASS  |

The archived “total aromatics” value was 7.3494 wt% and excluded PA. The inclusive value above is recomputed from its component mass record, adding PA consistently with the current total-aromatics duty. Neither value meets the 5 wt% target. Water and NMP are excluded from the hydrocarbon denominator.

## Hydraulic candidates—not final selections

| RPM | Diameter (m) | Hardware pitch (m) | Stored hydraulic-pass flag |
|-----|--------------|--------------------|----------------------------|
| 30  | 0.8687       | 0.4343             | False                      |
| 35  | 1.1988       | 0.5994             | True                       |
| 40  | 1.6470       | 0.8235             | True                       |
| 45  | 2.2722       | 1.1361             | False                      |
| 50  | 3.2030       | 1.6015             | False                      |

These are saved in-range hydraulic trials. Their stored pass flags are preserved literally; the table does not independently revalidate them or recommend a final RPM/diameter. Local operating hydrodynamics and transport must be recalculated and qualified for the ultimately selected column.

## Physical compartments and efficiency

The recorded hardware relation is compartment pitch = 0.5 × diameter. The seven finite-volume cells and ten configured equilibrium stages are **not physical compartments**. The current theoretical-stage authority is the explicitly labelled pre-pilot default  $N_T = 7$ , not a calculated  $N_T$ .

A physical integer count must satisfy the full duty at its exact hardware height. Efficiency is then calculated as  $N_T$ /physical count, retaining that theoretical-stage provenance—not assumed to obtain a height.

## Numerical closeout

The completed 33-node local trial converged, with full reduced rank 391/391 and nonnegative Bernstein film profiles. Its independent constitutive defects were 1.648e-8 (continuous) and 1.834e-7 (dispersed), above the unchanged 1e-8 limit. A later driver result-processing NameError stopped the attempt after 338.12 seconds; this was not budget exhaustion or a physical-instability finding. That processing error was fixed and three native-free regression tests passed, but no further physical run was performed. Two qualifying successive refinements, independent final-root reproduction, exact-candidate stability and whole-column sizing remain uncompleted.

## Sulfur remains a separate qualification

The archived allocation-based sulfur post-processing reported 1163.2 ppm against 1000 ppm. This is a conditional family-retention estimate, not independently species-resolved sulfur transport. Aromatic transfer alone does not qualify the sulfur duty.

**Decision boundary:** a qualified local film still needs a qualified whole-column calculation, exact hardware-height replay and all applicable product targets before any of the five requested outputs can be accepted. No production A/B/C reruns or saved-input changes were made.

## Evidence

prepilot-sizing-current-basis-db.json  
5e3223e8658f4424cd5d95ee3a7c22e07d9f6fdda7b33b73dc8cb9fb0d73931a

pre-pilot-sizing-equilibrium-duty-db.json  
cd4309f8a182a32ac3d540cac9fb262d956fde58e20e48711ac99551dd8a61a3

pre-pilot-sizing-authority-v1.readiness.json  
3c6476e63b3a5a240500f69e8e7d8757921021b45c6de3458d162ffb50c7d096

pre-pilot-sizing-numerical-closeout.json  
0f77c0634fef62730b113e66b721c9f7662e04ddeac238e0d5fcc07dff1ab2c